

A packing-dimension condition for positive pinned distances

The theorem, its meaning, and a focused proof

30 September 2026

1 The result

Fix a point y of a planar set E . Its *pinned distance set* is

$$\Delta_y(E) = \{|x - y| : x \in E\}.$$

The question is whether these distances occupy positive length on the real line. Write $d = \dim_{\text{H}} E$ and $D = \dim_{\text{P}} E$. Hausdorff dimension supplies a measure that does not concentrate too much in small balls; packing dimension controls the number of small squares needed after restricting to a suitable part of the set.

Theorem 1.1. *Let $E \subset \mathbb{R}^2$ be Borel. If*

$$1 < d = \dim_{\text{H}} E \leq \frac{5}{4}, \quad \dim_{\text{P}} E < B_{\text{H}}(d),$$

then there is a point $y \in E$ such that $\mathcal{L}^1(\Delta_y(E)) > 0$.

The sufficient boundary is

$$B_{\text{H}}(d) = \begin{cases} 2d - 1, & 1 < d \leq d_{\alpha}, \\ 1 + r_2(d - 1), & d_{\alpha} < d \leq d_c, \\ \frac{1}{3 - 2d}, & d_c < d \leq \frac{5}{4}, \end{cases} \quad (1)$$

where

$$d_{\alpha} = \frac{7 - \sqrt{7}}{4} \approx 1.088562172, \quad d_c = \frac{2 + \sqrt{6}}{4} \approx 1.112372436,$$

and the short middle branch is given explicitly by

$$r_2(a) = \frac{6a}{2 - a + 8a^2 + \sqrt{4 - 28a - 63a^2 + 32a^3 + 64a^4}}. \quad (2)$$

The radical is real throughout the required interval, and both joins are continuous; Appendix A proves these facts.

The packing inequality is *strict*. At $d = 5/4$ it permits $D < 2$, but does not include $D = 2$. For $d > 5/4$, the theorem of Guth–Iosevich–Ou–Wang already gives a positive-length pinned distance set without a packing restriction [3]. No necessity or global optimality claim is made for (1).

2 How to read the bound

For most of the improved range, the formula is simply $B_H(d) = 1/(3 - 2d)$. The radical only joins this branch to $2d - 1$ on the small interval between approximately 1.0886 and 1.1124.

Hausdorff dimension d	Sufficient packing restriction	An allowed value
1.05	$D < 1.10$	$D = 1.08$
1.10	$D < 28/23 \approx 1.21739$	$D = 1.21$
1.15	$D < 10/7 \approx 1.42857$	$D = 1.40$
1.20	$D < 5/3 \approx 1.66667$	$D = 1.60$
1.24	$D < 25/13 \approx 1.92308$	$D = 1.90$
1.25	$D < 2$	$D = 1.99$

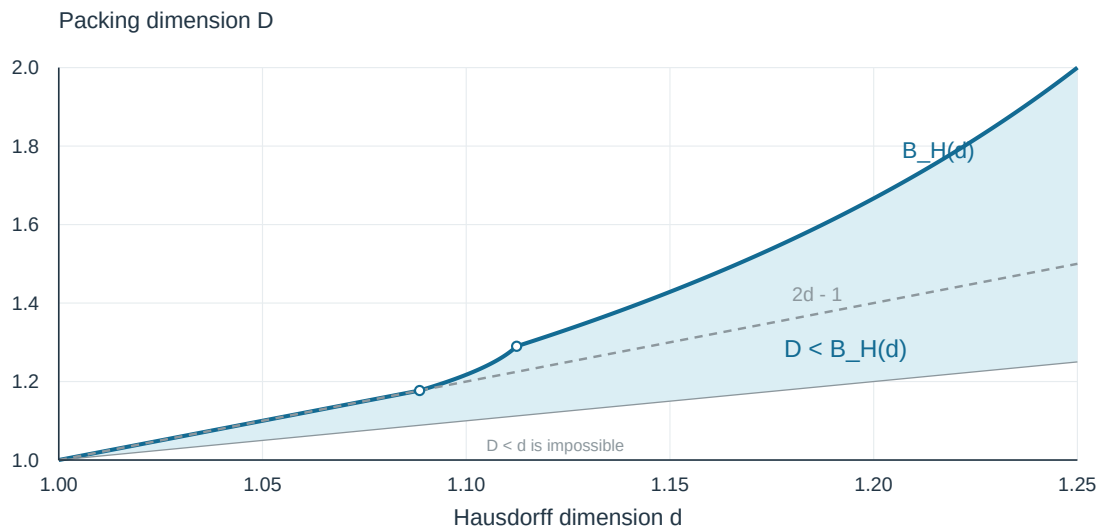


Figure 1: The shaded region $d \leq D < B_H(d)$ is covered by the theorem. Its upper boundary is excluded. The dashed line shows the simpler condition $D < 2d - 1$. The graph illustrates the formulas; it is not used as evidence for the inequalities.

For example, at $d = 1.24$, the simpler condition requires $D < 1.48$, whereas the theorem permits $D < 25/13$. The pin is a point of the original set, not a point chosen elsewhere in the plane.

3 The idea of the proof

Choose a probability μ on one compact part of E , and a probability ν on another, separated compact part. The distance distribution from a pin y is the pushforward

$$(d_y)_*\mu, \quad d_y(x) = |x - y|.$$

It suffices to prove that this probability has a Lebesgue density for ν -almost every pin. Such a density has integral one, so its supporting distance set cannot have zero length.

The proof supplies this density in two regimes. Under $D < 2d - 1$, replace distance by its tangent affine map in small source squares. The resulting positive densities converge in L^1 : their average absolute differences are summable. Outside that regime, split the source into frequency annuli.

In each annulus discard a small L^1 error and prove a summable L^2 bound for the remaining density.

The second argument is where the improved curve comes from. The pin measure's mass distribution across scales is encoded by a Lipschitz function. Rather than fixing the same intermediate scales for every distribution, choose a short chain of scales adapted to that function. A one-dimensional calculation bounds the total loss along the chain. If this loss is smaller than $d - 1$, the annular density estimates are summable.

Thus the proof has three separate tasks: extract compact measures, bound a scalar scale-selection problem, and turn that bound into a density. The remainder of the main text states their precise interfaces and assembles them. The appendices provide their proofs without the exploratory material from the longer research manuscript.

4 A precise proof of the theorem

4.1 Measures and compact reduction

A probability λ is t -Frostman when $\lambda(B(x, r)) \leq Cr^t$ for $0 < r \leq 1$. Define

$$I_s(\lambda) = \iint |x - x'|^{-s} d\lambda(x) d\lambda(x').$$

On a bounded support, the Frostman estimate implies $I_s(\lambda) < \infty$ for every $0 < s < t$, by summing the concentric distance annuli. Let $N(K, r)$ be the least number of radius- r balls covering K .

Lemma 4.1 (Compact measures with the required exponents). *If E is Borel, $0 < t < \dim_{\text{H}} E$, and $\dim_{\text{P}} E < u$, there are separated compact sets inside E carrying t -Frostman probabilities μ, ν such that each support has covering number at most Cr^{-u} .*

Proof. Frostman's lemma supplies a compactly supported t -Frostman probability on E . The countable-cover characterization of packing dimension covers E by bounded sets of upper box dimension strictly below u . Taking closures and intersecting with E makes this a Borel cover without changing the upper box bounds. One member has positive Frostman mass. Inner regularity gives a compact subset of positive mass in it; its covering numbers are bounded by Cr^{-u} . The restricted measure is nonatomic, so choose two disjoint small closed balls, a positive distance apart, whose intersections with that compact set have positive mass. Normalize the two restrictions. The Frostman and covering bounds persist, with new constants. $\square$

4.2 The two density estimates

Proposition 4.2 (The direct comparison). *Suppose separated compact probabilities μ, ν are t -Frostman, where $1 < s < \min\{t, 2\}$, and $N(\text{supp } \mu, r) \leq Cr^{-u}$. If $u < 2s - 1$, then $(d_y)_* \mu$ is absolutely continuous for ν -almost every y .*

The proof in Appendix C constructs the positive affine approximations described above and proves their L^1 convergence.

Here is the interface for the improved branch. For a 1-Lipschitz profile $g : [0, N] \rightarrow \mathbb{R}$, the cost of an edge from n down to $m < n$ is

$$c_g(m, n) = g(m) - \min_{m \leq x \leq n} g(x).$$

The edge is permitted when $2n - m \leq N$. Its cost records a possible loss caused by concentration between those two scales.

Proposition 4.3 (The profile-to-density implication). *Suppose separated compact probabilities μ, ν are t -Frostman, where $1 < s < t$, and $N(\text{supp } \nu, r) \leq Cr^{-u}$, with $s < u < 2$. Suppose there is a number $\mathcal{C} < s - 1$ with the following property.*

Every piecewise affine, 1-Lipschitz profile satisfying

$$g(0) = 0, \quad (s - 1)x \leq g(x) \leq (u - 1)x$$

has a permitted chain from $n_0 = (1 - \zeta)N$ to zero whose total cost is at most CN , allowing an arbitrarily small additional multiple of N . The number of edges is bounded in terms of $\zeta > 0$, independently of N and the profile. Rounding to a grid of mesh T and allowing additive barrier errors of size ϵN changes the bound by at most $CK(\epsilon N + T)$, where K is the chain length.

Then $(d_y)_\mu$ is absolutely continuous for ν -almost every pin.*

Appendix B proves this implication, including the finite measure regularization, deletion estimate, Fourier iteration, and identification of the limiting density. Appendix D proves the one-step Fourier estimate used there.

The real-variable calculation in Appendix A establishes exactly the following sufficient input:

$$1 < s \leq \frac{5}{4}, \quad 2s - 1 \leq u < B_H(s) \implies \text{the profile bound holds with some } \mathcal{C} < s - 1. \quad (3)$$

That calculation also proves continuity of B_H , so all exponent choices in the next paragraph are justified by strict inequalities.

4.3 Assembly

Proof of Theorem 1.1. If $D < 2d - 1$, choose $1 < s < t < d$ and $D < u < 2s - 1$. Lemma 4.1 gives the two compact probabilities, and Proposition 4.2 gives their pinned densities.

Otherwise $2d - 1 \leq D < B_H(d)$. Continuity and strictness allow $s < d$, sufficiently close to d , and then t, u such that

$$1 < s < t < d, \quad D < u < B_H(s), \quad 2s - 1 < u < 2.$$

When $d = 5/4$, choose $s < 5/4$; the strict condition $D < 2$ still provides the required room. Apply Lemma 4.1. Equation (3) and Proposition 4.3 give absolute continuity of $(d_y)_*\mu$ for ν -almost every pin.

In either case choose such a pin y in $\text{supp } \nu \subset E$. The probability $(d_y)_*\mu$ is supported on the compact set $\Delta_y(\text{supp } \mu) \subset \Delta_y(E)$. Absolute continuity and total mass one imply that this compact distance set has positive Lebesgue measure. This proves the theorem. $\square$

4.4 Established results used in the appendices

The proof uses Frostman's lemma and the packing-dimension covering characterization; Plancherel and the Riesz energy identity; Orponen's radial projection theorem in the averaged form of GIOW Theorem 3.7 or Keleti–Shmerkin Proposition 3.11; Keleti–Shmerkin Lemma 3.4 and Corollary 3.5 for finite regularization, and Lemma 5.4 for total drop; the standard packet localization in GIOW Section 3 and Liu's Lemma 4.2; and Liu's pinned quadratic identity. These inputs are cited where used. The new profile bounds and variable-chain Fourier estimate are proved here.

A Optimizing the scale profile

This section supplies the real-variable part of the proof. A profile records how the masses of the regularized pin measure change with scale. The analytic argument needs a decreasing sequence of scales with small total cost; the purpose here is to construct that sequence and evaluate the resulting dimension condition explicitly.

A.1 The cost to be bounded

Let $g : [0, N] \rightarrow \mathbb{R}$ be finitely piecewise affine, 1-Lipschitz, and satisfy

$$ax \leq g(x) \leq bx, \quad 0 < a < b \leq 1.$$

An edge from n to $m < n$ is admissible if $2n - m \leq N$. Its cost is

$$c_g(m, n) = g(m) - \min_{m \leq x \leq n} g(x). \quad (4)$$

Costs are nonnegative. No intermediate endpoint is prescribed.

The two bounds we shall use are

$$C_1(a, b) = \frac{b - a}{1 + 2b}, \quad (5)$$

$$C_2(a, b) = \frac{(b - a)(2 + a - 2b - 4ab)}{(1 - a)(1 + 2b)^2}. \quad (6)$$

Also put

$$b_c(a) = \frac{1 + 2a}{4 + 2a}, \quad R(a, b) = 4(1 + a)b^2 - 4(a + 2)b + 1 + 3a. \quad (7)$$

Theorem A.1 (The profile estimate). *Suppose $0 < a \leq 1/4$, $a < b \leq 1$, and let g satisfy the preceding hypotheses. For every $0 \leq n_0 < N$, an admissible chain from n_0 to zero exists with total cost at most $NC(a, b) + \rho N$, for every prescribed $\rho > 0$, where either of the following choices is valid:*

$$\begin{aligned} C(a, b) &= C_2(a, b), \quad b \leq b_c(a) \text{ and } R(a, b) \leq 0, \\ C(a, b) &= C_1(a, b), \quad b \geq b_c(a). \end{aligned}$$

The chain has at most

$$K \leq C \left(1 + \log \frac{N}{N - n_0} \right) \quad (8)$$

edges, with an absolute constant. If a grid of mesh T contains $0, n_0, N$, its endpoints can be chosen on that grid with additional cost $O(KT)$. If the profile barriers hold with uniform additive error εN , the additional cost is $O(K\varepsilon N + KT)$.

We first work on $[0, 1]$. The arbitrary accuracy ρ avoids any need to prove attainment of an infimum; the analytic argument has a strict exponent margin and will choose ρ smaller than that margin. The finite-chain and grid assertions are established at the end.

A.2 Normalizing the endpoint and identifying the relevant gaps

Replace g by

$$\tilde{g}(x) = \min\{g(x), 1 + a - x\}.$$

This is still 1-Lipschitz with the same two barriers, and $\tilde{g}(1) = a$. Furthermore

$$c_g(m, n) \leq c_{\tilde{g}}(m, n). \quad (9)$$

If $g(m) \leq 1 + a - m$, the left endpoint is unchanged and the minimum can only decrease. Otherwise the new cost is at least $(1 + a - m) - (1 + a - n) = n - m$, which bounds the old cost by Lipschitz continuity. We may therefore assume $g(1) = a$.

We use one published real-variable identity. In the notation below it is precisely the hard-point formula of Keleti–Shmerkin [2, Lemma 5.4]: if f is a piecewise affine 1-Lipschitz function with $f(0) = 0$, then its infimum of total drops over partitions

$$1 = t_0 > t_1 > \cdots \downarrow 0, \quad t_{i-1} \leq 2t_i,$$

where a drop on $[t_i, t_{i-1}]$ uses its lower endpoint, $f(t_i) - \min_{[t_i, t_{i-1}]} f$, equals the sum of $f(u) - f(v)$ over the interval components $[u, v]$ of

$$\{t : f(t) = \min_{[t/2, t]} f\}.$$

We now translate this identity carefully to the cost (4).

Let $\Phi_\zeta(g)$ be the infimum of chain costs from $1 - \zeta$ to zero, and set $f(t) = g(1 - t) - a$. The edge $n \rightarrow m$ corresponds to $x = 1 - n < y = 1 - m$, with $y \leq 2x$. Its cost is $f(y) - \min_{[x, y]} f$. The corresponding total-drop cost is $f(x) - \min_{[x, y]} f$. Their difference is $f(y) - f(x)$, so telescoping gives

$$\Phi_\zeta(g) = T_\zeta(f) - g(1 - \zeta),$$

where $T_\zeta(f)$ is the infimum of the usual total drop on $[\zeta, 1]$. Extending a finite partition below ζ by successive halvings adds at most ζ to the cost. Conversely, truncating an infinite partition at the interval crossing ζ changes that interval's cost by at most 2ζ ; the ratio condition is preserved. Hence

$$T(f) - \zeta \leq T_\zeta(f) \leq T(f) + 2\zeta, \quad \Phi_0(g) := \lim_{\zeta \downarrow 0} \Phi_\zeta(g) = T(f) - a.$$

In the original coordinate the hard-point set is

$$H_g = \{x : g(x) = \min_{[x, (1+x)/2]} g\}.$$

For a piecewise affine function this is a finite union of closed intervals, possibly including degenerate intervals. Both zero and one belong to it. The function g is nondecreasing on every component: hardness compares sufficiently close pairs in that interval, and subdivision compares arbitrary pairs. A component $[p, q]$ becomes $[1 - q, 1 - p]$ under reversal. The quoted identity therefore reads

$$\Phi_0(g) = \sum_{[p, q] \text{ component of } H_g} (g(q) - g(p)) - a. \quad (10)$$

The subtraction of a is essential.

Lemma A.2 (The geometry of a gap). *If (q, p) lies between consecutive components of H_g , then*

$$g(x) \geq g(p) \quad (q < x < p), \quad g(q) \geq g(p).$$

Its drop $\delta = g(q) - g(p)$ is nonnegative; if it is positive,

$$\delta \leq p - \frac{1 + q}{2}. \quad (11)$$

Proof. Suppose $g(x) < g(p)$ for some interior point. A minimum point z of g on $[x, p]$ lies before p and has value below $g(p)$. On $[z, p]$ the profile is at least $g(z)$. Any remaining part of $[z, (1 + z)/2]$ lies in $[p, (1 + p)/2]$, where hardness of p gives $g \geq g(p) > g(z)$. Thus z would be hard, contradicting the choice of the gap. Continuity proves the endpoint inequality. If the drop is positive, hardness of q forces $p > (1 + q)/2$ and $g((1 + q)/2) \geq g(q)$. Lipschitz continuity from $(1 + q)/2$ to p gives (11). $\square$

Telescoping the changes along all components and gaps, using $g(0) = 0$, $g(1) = a$, turns (10) into

$$\Phi_0(g) = \sum_{\text{gaps}} \delta. \quad (12)$$

We may discard gaps with zero drop.

Lemma A.3 (Controlling the remaining gaps). *If $g(p) = v$, the sum S of drops of all positive gaps contained in $[p, 1]$ satisfies*

$$S \leq \frac{1-p}{2}, \quad S \leq \frac{1-p+v-a}{2}.$$

Consequently, for $0 \leq \chi \leq 1/2$,

$$S \leq \frac{1-p}{2} + \chi(v-a). \quad (13)$$

Proof. A gap (q_i, p_i) has

$$\delta_i \leq p_i - \frac{1+q_i}{2} \leq \frac{p_i - q_i}{2}.$$

The intervals are disjoint, proving the first estimate. Independently, the drop is at most the negative variation of g on that gap. On $[p, 1]$, total variation is at most $1-p$, and the net change is $a-v$. Its negative variation is therefore at most $(1-p+v-a)/2$. A convex combination of the two estimates with weights $1-2\chi$ and 2χ proves the last statement. $\square$

If there is only one positive gap, the barriers and (11) give

$$\delta \leq p - \frac{1+q}{2}, \quad \delta \leq bq - ap.$$

Multiplying the first inequality by $2b$, adding the second, and dividing by $1+2b$ yields

$$\delta \leq \frac{(2b-a)p-b}{1+2b} \leq C_1(a, b). \quad (14)$$

With no positive gaps the cost is zero. We henceforth consider the first two positive gaps. Write

$$(q_1, p_1), (q_2, p_2), \quad u_i = g(q_i), \quad v_i = g(p_i), \quad \delta_i = u_i - v_i.$$

The remaining gaps will be handled by Lemma A.3.

A.3 The bound when b is small

Assume $b \leq b_c(a)$ and $R(a, b) \leq 0$. In particular $b < 1/2$. Put

$$s = 1 + 2b, \quad Z = (1-a)s^2, \quad \chi = \frac{1+a-4ab+4b^2-2b}{Z}.$$

The numerator is $1-2b+a+4b(b-a) > 0$, and

$$\chi - \frac{1}{2} = \frac{R(a, b)}{2Z} \leq 0.$$

Thus the blended tail estimate is legal.

The following seven inequalities follow respectively from the upper barrier, lower barrier, gap bound, Lipschitz continuity, upper barrier, gap bound, and the interval endpoint:

$$\begin{aligned} u_1 - bq_1 &\leq 0, \\ ap_1 - v_1 &\leq 0, \\ q_1 - 2p_1 + 2\delta_1 &\leq -1, \\ u_2 - v_1 - q_2 + p_1 &\leq 0, \\ u_2 - bq_2 &\leq 0, \\ q_2 - 2p_2 + 2\delta_2 &\leq -1, \\ p_2 &\leq 1. \end{aligned}$$

Multiply them, in that order, by

$$\begin{aligned} w_1 &= \frac{1}{s}, & w_2 &= \frac{1-2b}{(1-a)s}, & w_3 &= \frac{b}{s}, & w_4 &= \frac{2b-a}{(1-a)s}, \\ w_5 &= \frac{1+2a-4b-2ab}{Z}, & w_6 &= \frac{1-\chi}{2}, & w_7 &= \frac{1}{2} - \chi. \end{aligned}$$

Every multiplier is nonnegative: the sign of w_5 is exactly $b \leq b_c(a)$, and the other signs follow from $a < b < 1/2$ and $0 < \chi \leq 1/2$. For transparency, the cancellation uses

$$\begin{aligned} w_3 &= bw_1, & w_1 + 2w_3 &= 1, & aw_2 - 2w_3 + w_4 &= 0, & w_2 + 2w_3 + w_4 &= 1, \\ w_6 &= w_4 + bw_5, & w_4 + w_5 &= \chi, & 2w_6 &= 1 - \chi. \end{aligned}$$

The resulting inequality is

$$\delta_1 + \delta_2 - \frac{p_2}{2} + \chi v_2 \leq -w_3 - w_6 + w_7 = -\frac{b}{1+2b} - \frac{\chi}{2}.$$

Apply (13) after p_2 . Then

$$\begin{aligned} \Phi_0(g) &\leq \delta_1 + \delta_2 + \frac{1-p_2}{2} + \chi(v_2 - a) \\ &\leq \frac{1}{2} - \frac{b}{1+2b} - \left(a + \frac{1}{2}\right)\chi \\ &= C_2(a, b). \end{aligned}$$

The zero- and one-gap cases also satisfy this estimate because

$$C_2(a, b) - C_1(a, b) = \frac{(b-a)(1+2a-4b-2ab)}{(1-a)(1+2b)^2} \geq 0.$$

A.4 The bound for $b_c(a) \leq b \leq 1/2$

Between p_1 and q_2 , all hard components increase and all omitted gaps have zero drop. Hence $v_1 \leq u_2$. Use the eight inequalities

$$\begin{aligned} u_1 - bq_1 &\leq 0, \\ \delta_1 - p_1 + q_1/2 &\leq -1/2, \\ ap_1 - v_1 &\leq 0, \\ \delta_2 - p_2 + q_2/2 &\leq -1/2, \\ ap_2 - v_2 &\leq 0, \\ u_2 - v_1 - q_2 + p_1 &\leq 0, \\ p_2 &\leq 1, \\ v_1 - u_2 &\leq 0. \end{aligned}$$

Let $Z_1 = (1 + 2a)(1 + 2b)$. Their respective multipliers are

$$\begin{aligned}\omega_1 &= \frac{1}{1 + 2b}, & \omega_2 &= \frac{2b}{1 + 2b}, \\ \omega_3 &= \frac{2b}{Z_1}, & \omega_4 &= \frac{4b(1 + a)}{Z_1}, \\ \omega_5 &= \frac{1 + 2a - 2b}{Z_1}, & \omega_6 &= \frac{2b(1 + a)}{Z_1}, \\ \omega_7 &= \frac{(6 + 8a)b - (1 + 2a)^2}{2Z_1}, & \omega_8 &= \frac{(2a + 4)b - (2a + 1)}{Z_1}.\end{aligned}$$

All are nonnegative. The last sign is $b \geq b_c(a)$. The numerator of ω_7 increases with b ; at $b = b_c(a)$ it is

$$\frac{(1 + a)(1 - 2a)(1 + 2a)}{a + 2} > 0.$$

The remaining signs follow directly from $0 < a \leq 1/4$ and $b \leq 1/2$.

Expanding the weighted sum gives

$$\delta_1 + \delta_2 - \frac{p_2}{2} \leq -\frac{\omega_2}{2} - \frac{\omega_4}{2} + \omega_7.$$

Using the first tail estimate after p_2 , we obtain

$$\Phi_0(g) \leq \frac{1}{2} - \frac{\omega_2}{2} - \frac{\omega_4}{2} + \omega_7 = C_1(a, b).$$

The zero- and one-gap cases follow from (14).

A.5 The bound for $b \geq 1/2$

Here a direct potential is shorter than a gap calculation. Define

$$L(x) = 2bx - b - g(x), \quad V(x) = \frac{\max\{L(x), 0\}}{1 + 2b}.$$

The function L is nondecreasing, since $g' \leq 1 \leq 2b$ almost everywhere. Moreover

$$(-g')_+ \leq \frac{2b - g'}{1 + 2b}.$$

For $g' \geq 0$ the right side is nonnegative; for $g' < 0$, the inequality reduces to $g' \geq -1$. Every edge cost is at most the negative variation of g on its interval.

Above the zero region of L , take maximal admissible downward jumps, shortening the last one to reach that region. Their costs sum to at most $V(n_0)$. Once $L(n) \leq 0$, if $n > 1/2$,

$$g(2n - 1) \leq b(2n - 1) \leq g(n).$$

A minimum of g on $[2n - 1, n]$, chosen leftmost, therefore gives a zero-cost admissible continuation. When $n \leq 1/2$, the edge to zero is admissible and costs zero because $g(0) = 0 \leq g$.

To make termination precise, perform this construction on a uniform grid and a piecewise affine interpolation. In the positive- L region stop at the first grid point above its zero region and cross to the next point below, at cost at most the mesh. The zero-cost minima then occur at grid points and strictly decrease the endpoint, so the procedure terminates. If the prescribed start is not on this grid, an initial step to the nearest point below it costs at most the mesh and is admissible once the mesh is small. Thus the grid construction costs at most $V(n_0) + O(\text{mesh})$.

Compress the chain by the procedure below. Its length is then bounded independently of the mesh, so transferring it from the interpolant to g costs only $O(K \text{ mesh})$. Taking the mesh sufficiently small makes the total error less than the prescribed ρ . Since

$$V(n_0) \leq V(1) = \frac{b-a}{1+2b} = C_1(a, b),$$

this proves the required high- b estimate.

A.6 Finite chains and grid errors

First, an infimum for a finite start is bounded by the limiting cost used above. Truncating a chain from a larger starting point at its first crossing of a smaller start does not increase its cost: the crossing edge is replaced by a subinterval with the same lower endpoint, whose minimum can only increase. Conversely, maximal admissible edges bridge the two starting points at total cost at most their distance, by the variation bound. Hence

$$0 \leq \Phi_0(g) - \Phi_\zeta(g) \leq \zeta.$$

Every finite chain can be compressed without increasing its cost. For $l < m < n$,

$$c_g(l, n) \leq c_g(l, m) + c_g(m, n).$$

Indeed, if $A = \min_{[l, m]} g$ and $B = \min_{[m, n]} g$, the difference between the right side and left side is $g(m) - \max(A, B) \geq 0$. Merge neighboring edges whenever their union remains admissible. In the resulting chain, every pair of successive edges more than doubles the complementary distance from the endpoint one:

$$2n_i - n_{i+2} > 1 \implies 1 - n_{i+2} > 2(1 - n_i).$$

A start at $1 - \zeta$ therefore needs at most $2\lceil \log_2(1/\zeta) \rceil + O(1)$ edges.

Choose a finite chain within ρ of its finite-start infimum and compress it. The resulting cost is at most $C(a, b) + \rho$ and its length has the displayed bound. No limiting chain or attainment is needed. The high- b construction gives the same conclusion: its interpolant differs uniformly from g by at most the mesh, so each retained edge cost differs by at most twice the mesh. Rescaling gives (8) and the asserted cost $NC(a, b) + \rho N$.

Finally round all endpoints downward to a grid of mesh T containing $0, n_0, N$. Admissibility is preserved because

$$\left\lfloor \frac{m}{T} \right\rfloor \geq \left\lfloor \frac{2n - N}{T} \right\rfloor \geq 2 \left\lfloor \frac{n}{T} \right\rfloor - \frac{N}{T}.$$

Each endpoint moves at most T , so each cost changes by at most $2T$. If the original barriers have additive error εN , first clamp the profile between ax and bx . The result is still 1-Lipschitz and differs by at most εN ; a cost changes by at most twice that amount. Endpoint normalization only increases the cost available to the original profile. Summing these errors proves the final assertions of Theorem A.1.

A.7 Evaluating the dimension cutoff

Set

$$\alpha = \frac{3 - \sqrt{7}}{4}, \quad a_c = \frac{\sqrt{6} - 2}{4}, \quad b_F(a) = \frac{2a}{1 - 2a}.$$

For $\alpha \leq a \leq a_c$, define

$$r_2(a) = \frac{8a^2 - a + 2 - \sqrt{64a^4 + 32a^3 - 63a^2 - 28a + 4}}{4(1 + 4a - 2a^2)}. \quad (15)$$

The radical is real on this interval, as verified below. Rationalizing the numerator gives the expression in (2): if $A = 2 - a + 8a^2$ and $\Delta = 4 - 28a - 63a^2 + 32a^3 + 64a^4$, then $A^2 - \Delta = 24a(1 + 4a - 2a^2)$. The final curve is

$$B_H(d) = \begin{cases} 2d - 1, & 1 < d \leq 1 + \alpha, \\ 1 + r_2(d - 1), & 1 + \alpha < d \leq 1 + a_c, \\ \frac{1}{3 - 2d}, & 1 + a_c < d \leq 5/4. \end{cases} \quad (16)$$

Proposition A.4 (What the inequality supplies). *Let $1 < d \leq 5/4$ and $d \leq D < B_H(d)$. Then either*

$$D < 2d - 1,$$

or there are $s < d$, $u > D$, with $1 < s < 5/4$, $s < u < 2$, for which one of the two parameter conditions in Theorem A.1 holds and $C(s - 1, u - 1) < s - 1$.

Proof. Put $a = d - 1$, $b = D - 1$. Only $b \geq 2a$ needs examination. First let $\alpha \leq a \leq a_c$ and $2a \leq b \leq b_c(a)$. The guard $R(a, b) < 0$ holds throughout this region. Indeed

$$\partial_b R = 8(1 + a)b - 4(a + 2) \leq -4 \quad (b \leq 1/2),$$

and

$$R(a, 2a) = 16a^3 + 8a^2 - 13a + 1.$$

Its derivative is $48a^2 + 16a - 13 \leq -6$ on $[0, 1/4]$, whereas

$$R(\alpha, 2\alpha) = 3(11\alpha - 1) < 0.$$

For the last sign, $\alpha < 1/11$ is equivalent to $29 < 11\sqrt{7}$, and squaring gives $841 < 847$.

The derivative of C_2 in b is

$$\partial_b C_2 = \frac{2 + 11a + 8a^2 - (8 + 14a + 8a^2)b}{(1 - a)(1 + 2b)^3} > 0 \quad (b \leq b_c(a)).$$

The numerator decreases with b ; its value at $b_c(a)$, multiplied by $4 + 2a$, is $18a(1 + a) > 0$. At the lower endpoint,

$$C_2(a, 2a) - a = \frac{a(1 + 2a)(8a^2 - 12a + 1)}{(1 - a)(1 + 4a)^2}.$$

This is zero at $a = \alpha$ and negative for $\alpha < a \leq a_c$. At $b = b_c(a)$, the formulas C_1, C_2 agree. Also

$$C_1(a, b) < a \iff b < b_F(a).$$

The equality $b_F(a) = b_c(a)$ is equivalent to $8a^2 + 8a - 1 = 0$, hence to $a = a_c$. For $a < a_c$, $b_F(a) < b_c(a)$, so $C_2(a, b_c(a)) > a$.

There is therefore one root of $C_2(a, b) = a$ in $[2a, b_c(a)]$, with the indicated endpoint equalities. Clearing its positive denominator gives

$$(4a^2 - 8a - 2)b^2 + (8a^2 - a + 2)b - 3a = 0.$$

The smaller root is (15). The crossing has strictly positive derivative, so its discriminant is positive. This proves the reality and continuity of the radical expression. Its endpoint values are

$$r_2(\alpha) = 2\alpha, \quad r_2(a_c) = b_c(a_c) = \frac{\sqrt{6} - 1}{5}.$$

Thus the middle branch gives exactly the required strict cost inequality above the coherent range, and the curve is continuous at both joins.

Now let $a_c \leq a \leq 1/4$. For $b \geq b_c(a)$, Theorem A.1 gives $C_1 < a$ exactly when $b < b_F(a)$. If $b < b_c(a)$, increase the upper profile barrier to $b_c(a)x$. For $a > a_c$, the preceding crossing calculation gives $b_c(a) < b_F(a)$, so the high-side estimate at that larger barrier has

$$C_1(a, b_c(a)) < a.$$

At $a = a_c$ and $b < b_c(a)$, the strict low-side estimate has already been proved in the middle branch. Finally $1 + b_F(a) = 1/(1 - 2a) = 1/(3 - 2d)$, proving the last branch of (16).

All inequalities used by the profile branch are strict in cost. The guard is also strict wherever the low branch is used here. Away from the common boundary, continuity permits a small decrease of a and increase of the chosen upper barrier. At the common boundary one may move to its high side, since $C_1 = C_2$ there. In the enlarged-barrier case $a > a_c$, $b < b_c(a)$, choose $a' < a$ sufficiently close that $a' > a_c$ and $b_c(a') > b$, and use the upper barrier $b_c(a')$. These choices give $s = 1 + a' < d$ and some $u > D$ with the required strict profile bound. They may be chosen with $1 < s < 5/4$, $s < u < 2$, proving the assertion. $\square$

The analytic proof can now use the coherent implication when $D < 2d - 1$. In every remaining case it uses Theorem A.1 with the strict parameter slack provided by Proposition A.4. This is the complete profile input needed for the curve (16).

B From a scale profile to a distance density

This appendix proves Proposition 4.3. The strategy is to split each frequency annulus into a small discarded part and a retained part with a summable quadratic norm. Regularizing the pin measure lets its spatial mass distribution be described by one real-valued profile on each component. The chain cost is exactly the exponent paid by the retained Fourier estimate.

There are four steps: finite regularization; deletion of concentrated tubes; the Fourier estimate along the chosen chain; and summation of the resulting densities. The selected-cap estimate used in the third step is proved separately in Appendix D.

B.1 Finite regularization of the original pin measure

Here $\text{RapDec}(R)$ denotes a quantity bounded by $C_M R^{-M}$ for every fixed $M > 0$, uniformly over the cubes, caps and regularized components under discussion. All auxiliary small parameters are fixed before the frequency tends to infinity.

The constants in losses $R^{C\kappa^h}$ below are independent of the regularization block size T ; fixed annular ratios depending on T change only multiplicative constants.

Let μ, ν satisfy Proposition 4.3. Normalize the separated compact supports into the interior of one unit dyadic square by one common translation and dilation. Let $\mathcal{C} < s - 1$ be slightly larger than the profile cost supplied by that proposition's hypothesis, while still below $s - 1$. Fix the arbitrary accuracy in the profile bound smaller than this increase. Thus the additional multiple of N is absorbed into $\mathcal{C}N$ throughout the proof. Write

$$N(\text{supp } \nu, r) \leq Cr^{-u} \quad (0 < r \leq 1). \quad (17)$$

We shall prove

$$(d_y)_* \mu \ll \mathcal{L}^1 \quad \text{for } \nu\text{-almost every } y. \quad (18)$$

Constants can depend on these fixed measures and their separation. The radial-projection theorem of Orponen, in the form of GIOW Theorem 3.7 or Keleti–Shmerkin Proposition 3.11, supplies $q > 1$ such that

$$B = \int \|\pi_*^y \mu\|_{L^q(S^1)}^q d\nu(y) < \infty. \quad (19)$$

Here $\pi^y(x) = (x - y)/|x - y|$. We also have $I_s(\mu) < \infty$.

Fix a large integer T . Use a smooth partition of frequency space into a low-frequency ball and annuli indexed by $R = 2^N$, $N = 4T\ell$. The annular ratios depend only on T ; all localization constants below may do so as well. Write P_R for the annular multiplier, including a fixed smooth spatial cutoff equal to one near $\text{supp } \mu$ and supported a positive distance from $\text{supp } \nu$. These multipliers have uniformly bounded convolution-kernel first norms, with constants depending on T .

Apply Keleti–Shmerkin's finite regularization, Lemma 3.4 and Corollary 3.5, at depth N . More precisely, first distribute each depth- N cube mass uniformly within that cube. The lemma partitions all but mass $2^{-\epsilon N}$ into disjoint unions X_i of terminal cubes. Retain the *original* restrictions

$$w_i = \nu(X_i), \quad \sigma_i = w_i^{-1} \nu|_{X_i}.$$

They have the same coarser cube masses as the regularized measures. With

$$\Gamma = \epsilon + \frac{\log_2(4T + 2) + 2}{T},$$

we have $w_i \geq R^{-\Gamma}$. For each i , a function f_i , linear on the T -grid with slopes in $[0, 2]$, satisfies $f_i(0) = 0$ and, for every occupied cube of block depth k ,

$$2^{-f_i(k)-N/T} \leq \sigma_i(Q) \leq 2^{-f_i(k)}. \quad (20)$$

This follows directly by multiplying the one-step mass ratios in the definition of a regular measure. Also

$$sk - CTN - C_T \leq f_i(k) \leq uk + C_T. \quad (21)$$

For the first inequality use $\sigma_i \leq R^\Gamma \nu$, the Frostman bound, and the lower estimate in (20). For the second, the upper estimate in (20) forces at least $2^{f_i(k)}$ occupied cubes, whereas (17) bounds their number by $C2^{uk}$. Interpolation extends the barriers with $O(T)$ error. Finally,

$$\int \|\pi_*^y \mu\|_q^q d\sigma_i(y) \leq BR^\Gamma. \quad (22)$$

Put $g_i = f_i - \text{id}$. Apply the profile-chain hypothesis with $a = s - 1$, $b = u - 1$. Start at the T -grid point immediately below $(1 - \zeta)N$, where $0 < \zeta < 1/8$ is fixed. We obtain a chain, common to the entire component,

$$n_0 > n_1 > \cdots > n_K = 0, \quad 2n_j - n_{j+1} \leq N, \quad (23)$$

$$K \leq K_0(\zeta), \quad \sum_{j < K} c_{i,j} \leq CN + CKTN + C_{T,K}, \quad (24)$$

$$c_{i,j} = g_i(n_{j+1}) - \min_{[n_{j+1}, n_j]} g_i.$$

No intermediate endpoint is prescribed. Admissibility of the final edge implies $n_{K-1} \leq N/2$, which is all that the final Fourier estimate will need. Grid interpolation in the clamping step only changes the last constant. The source μ is never restricted or regularized in this process.

B.2 Truncated conditional energies and deletion

For one edge write $m = n_{j+1}$, $n = n_j$, $b = 2^{-m}$, $a = 2^{-n}$, and $\rho = a/b$. A cube is called active when its σ_i -mass is positive. Only active cubes and their ancestors occur below.

Let Q be an active depth- m cube. For any concentric enlargement Q^+ by a factor between one and $R^{C_K h}$, set $\lambda = (\sigma_i)_{Q^+}$. For $m \leq k \leq n$, covering a ball by boundedly many dyadic cubes and using (20) gives

$$\lambda(B(z, 2^{-k})) \leq CR^{CT} 2^{f_i(m) - f_i(k)}.$$

Indeed, the denominator is at least the mass of the original active cube Q ; this avoids any boundary-sliver assumption about Q^+ . For intermediate radii between consecutive block-grid points, use the mass bound at the coarser grid point. The ratio between adjacent radii is at most 2^T , so this changes the estimate by a multiplicative constant depending only on T (one may take $O(2^{2T})$). It introduces no additional power of R . Summing dyadic annuli yields

$$J_\lambda(\rho) := \iint \min\{\rho^{-1}, b/|y - z|\} d\lambda(y) d\lambda(z) \leq C(N + 1)R^{CT} 2^{c_{i,j}}. \quad (25)$$

For separations larger than b , use the bound one. For the other annuli the summand is at most

$$CR^{CT} 2^{k-m+f_i(m)-f_i(k)} = CR^{CT} 2^{g_i(m)-g_i(k)}.$$

The innermost ball satisfies the same estimate. Replacing b , the tube width, or the truncation parameter by factors at most $R^{C_K h}$ changes (25) by at most another factor $R^{C_K h}$. This follows directly from the minimum kernel; no infinite conditional energy is used.

Here is the deletion argument with its multiplier exposed. If a probability λ is in a ball of radius b , take a ρ -net of unoriented directions and, in each direction, a bounded-overlap cover by ρb -wide tubes. Pair counting gives

$$\sum_T \lambda(T)^2 \leq C J_\lambda(\rho). \quad (26)$$

For fixed y, z , at most $C \min\{\rho^{-1}, b/|y-z|\}$ directions can place both points in such a tube; the spatial tube multiplicity in one direction is bounded. Integrating this statement proves (26). Enlarged supports and tubes introduce only powers of the enlargement factor.

For each pin, delete source directions whose angular maximal density exceeds a threshold L . The maximal inequality on the circle and Holder's inequality imply

$$(\mu \times \sigma_i)(\text{source-heavy pairs}) \leq C_q L^{1-q} B R^\Gamma. \quad (27)$$

For completeness, if v_y is the angular density, then $\int_{\{M v_y > L\}} v_y \leq C_q L^{1-q} \|v_y\|_q^q$: bound the length of that set by $C_q L^{-q} \|v_y\|_q^q$, then use Holder. Intervals centered at a retained direction therefore have mass at most a constant times their length times L .

Now declare a pin tube heavy when $\lambda(T) > H\rho$. Associate to it the unit-length source tube in the same direction, widened to angular width $C\rho$. If this tube contains any pair not source-heavy, its source mass is at most $CL\rho$. The fixed source-pin separation justifies this correspondence. Consequently the retained mass of pairs using heavy pin tubes is at most

$$CL\rho \sum_{\lambda(T) > H\rho} \lambda(T) \leq \frac{CL}{H} \sum_T \lambda(T)^2 \leq \frac{CL}{H} J_\lambda(\rho). \quad (28)$$

One can discretize all tube centers and directions here: replacing a tube by a containing member of a grid changes widths by fixed factors. Directions within $O(\rho)$ have the same containing tube after this enlargement. Thus the estimate also bounds the union over the finer angular caps used below, without paying their number.

Sum (28) over active parents with weights $\sigma_i(Q^+)$. Concentric enlarged cubes at a fixed scale have overlap at most the square of their dilation factor. Equations (25)–(28) therefore remain valid with an additional $R^{C_K h}$. Choose

$$L = R^{D_1 h}, \quad H_{i,j} = R^{D_2 h + C\Gamma} 2^{c_{i,j}}. \quad (29)$$

Here D_1 is sufficiently large in terms of q, K , and D_2 is sufficiently large compared with D_1, K . Then, for Γ sufficiently small compared with h , the deleted pair mass is at most $CR^{-c_1 h}$. The constant c_1 can be chosen larger than any fixed geometric overlap exponent needed below, by increasing D_1, D_2 . If $H_{i,j}\rho \geq 1$, there are no heavy pin tubes. Factors caused by an enlargement are covered by increasing D_1, D_2 ; their dependence on K, q is permitted.

B.3 Fourier iteration with no prescribed midpoint

We use the regularized component σ_i , the conditional energy estimate (25), and the deletion estimates (27)–(28). The chain required in this subsection has the following properties:

$$\begin{aligned} n_0 > n_1 > \cdots > n_K = 0, \quad 2n_j - n_{j+1} \leq N, \quad n_0 = (1 - \zeta)N + O(T), \quad (30) \\ K \leq K_0, \quad \sum_{j < K} c_{i,j} \leq \mathcal{C}N + CK\Gamma N + C_{T,K}, \quad c_{i,j} = g_i(n_{j+1}) - \min_{[n_{j+1}, n_j]} g_i. \end{aligned}$$

Here $0 < \zeta < 1/8$, K_0 is fixed independently of $R = 2^N$, and the depths lie on the T -grid. Here $\mathcal{C}$ is the cost bound in Proposition 4.3. No intermediate depth is prescribed. We retain $N \in 4T\mathbb{N}$, so the standard angular scale used below is dyadic.

Put

$$a_j = 2^{-n_j}, \quad r_j = Ra_j, \quad \delta_j = r_j^{-1}, \quad \rho_j = \frac{a_j}{a_{j+1}}, \quad \delta_* = R^{-1/2}.$$

The two elementary inequalities

$$\delta_j \leq \rho_j, \quad \delta_* \leq \rho_j \quad (0 \leq j < K) \quad (31)$$

have different roles. The first is equivalent to $r_{j+1} \leq r_j^2$. For the second, admissibility and $n_{j+1} \geq 0$ give

$$n_j - n_{j+1} \leq \min\{n_j, N - n_j\} \leq N/2.$$

The last edge also gives $n_{K-1} \leq N/2$, hence $\delta_{K-1} \leq \delta_*$. The second inequality in (31) permits the standard source wave packets to be retained even when the auxiliary Fourier caps become narrower than δ_* .

A fixed source packet decomposition and a separate label tree

Fix a smooth annular multiplier at the scale R . First partition it into the usual smooth standard-cap multipliers $\psi_{R,S}$, with angular width δ_* . The label S has a nominal dyadic angular arc of that length; its multiplier is supported in a fixed enlargement of the arc, and these supports have bounded overlap. These standard multipliers are fixed before any finer partition is made.

Use their usual source wave packets, with spatial tubes of width $R^{-1/2+h}$ and fixed bounded length. Choose the longitudinal span to cover the common normalized source and pin region, with its cutoff outside that region. Thus remoteness there is transverse separation, not an artifact of a tube endpoint. Write

$$M_T \mu = \chi \eta_T (\psi_{R,S(T)} \hat{\mu})^\vee.$$

Here χ is a fixed smooth compact source cutoff, equal to one on a neighborhood of $\text{supp } \mu$, and with support separated from the pin set. For each standard label the smooth spatial functions η_T form a partition of unity on $\text{supp } \chi$. Thus summing all standard packets gives precisely $P_R \mu$, with the source cutoff convention used above. All packet localization estimates are applied only to these smooth standard multipliers and these standard spatial tubes.

We now define a separate angular tree for Fourier estimates. For each level with $\delta_j \geq \delta_*$, its labels group whole standard labels according to their nominal dyadic ancestors of width δ_j . Their multipliers are sums of the corresponding $\psi_{R,S}$. For $\delta_j < \delta_*$, a level label is a pair (S, I) , where I is a dyadic angular interval of length δ_j meeting the support of $\psi_{R,S}$; its multiplier is

$$\psi_{R,S}(\xi) \mathbf{1}_I(\xi/|\xi|).$$

The intervals partition the entire angular support of the standard multiplier, including the part outside its nominal arc. Consequently each such interval is within angular distance $O(\delta_*)$ of the standard label's direction. Use half-open angular intervals to specify the partition at endpoints. The endpoint choices do not affect any circular integral.

After the standard scale, parents keep the same label S and group the finer dyadic intervals. Before it, parents group whole standard labels. If an edge of the spatial chain crosses the standard angular scale, insert that scale in the angular tree only. This creates no additional spatial level and no additional inflation factor. The notation $\theta \prec \alpha$ below means descent in

this label tree. It is a relation between labels; overlapping smooth standard supports cause no ambiguity in this relation.

Every level- j multiplier has angular support of width $O(\delta_j)$, and the supports have bounded overlap at each level. For fine labels this follows because at each direction only boundedly many standard multipliers are nonzero, and the dyadic intervals partition that direction separately for each such multiplier. For coarse labels it follows by grouping adjacent standard labels. No differentiability of a fine angular indicator will be used: these indicators partition spectral measures on circles, not source packets.

Coarsened marks and their bad-pair estimate

For a level- j Fourier label θ , define its effective label to be θ itself if $\delta_j \geq \delta_*$, and its standard parent S otherwise. Its angular width is

$$\bar{\delta}_j = \max\{\delta_j, \delta_*\} \leq \rho_j.$$

For each active a_j -cube $Q_j \subset Q_{j+1}$, mark an effective label bad when the enlarged a_j -by- a_{j+1} test tube, centered at Q_j and directed along that effective label, has mass greater than $H_{i,j}\rho_j$ under $(\sigma_i)_{Q_{j+1}^+}$. All Fourier labels with the same effective label receive the same mark. Descendant spatial cubes inherit the mark from their unique level- j ancestor. Use the thresholds (29).

For definiteness, the averaging cubes and test dilations may be chosen as

$$Q_j^+ = R^{(2j+2)h}Q_j, \quad \text{test-tube dilation } R^{(4K_0+20)h}.$$

For large R , child averaging cubes and their further R^h localizations fit inside the corresponding enlarged parent. Additional fixed geometric factors are absorbed into these dilations or their constants. The same enlarged parent is used in the marking condition, the truncated conditional energy, and the Fourier iteration.

These marks are unions of whole standard source-cap labels. At coarse levels this is true by their definition as groups of standard labels; at fine levels all descendants of S share the mark of S . For an initial pin cube Q_0 , remove a standard source packet when its enlarged tube $2T$ meets Q_0 and its standard label has any inherited bad mark. Call the retained source function G_{R,i,Q_0} .

The deletion estimates already proved apply to these marks. Indeed, directions within $O(\bar{\delta}_j)$ of an effective direction are within $O(\rho_j)$, so the same ρ_j -net of directions and the same containing tube grid cover the marked tubes. Their widths and conditional mass normalization are exactly those used in (25)–(28), up to the permitted enlargements. Source-heavy directions remain only an auxiliary device for estimating the bad mass; they are not additional Fourier marks.

The standard packet estimates give $\|M_T\mu\|_1 \lesssim \mu(CT) + \text{RapDec}(R)$, and rapidly small pinned first norm outside an enlarged T . A separated source-pin pair lies in only $R^{O(h)}$ enlarged standard packet tubes: their directions are within $O(R^{-1/2+h})$ of the connecting line, and the standard directions have spacing comparable to $R^{-1/2}$. Spatial tube multiplicity for a fixed direction is bounded. When a removed packet contributes to such a pair, its standard label and its effective bad label differ in direction by $O(\bar{\delta}_j)$; its packet uncertainty is $O(R^h\delta_*)$. By (31), both fit inside the enlarged conditional test tube based on that pair.

It follows, with $\text{Bad}_{R,i}$ the union of bad pairs estimated by the deletion argument, that

$$\int \|(d_y)_*(P_R\mu - G_{R,i,Q_0}(y))\|_1 d\sigma_i(y) \leq CR^{C_{K_0}h}(\mu \times \sigma_i)(\text{Bad}_{R,i}) + \text{RapDec}(R). \quad (32)$$

Only standard packets were summed here. There is no multiplicity equal to the number of auxiliary fine Fourier labels in a standard cap. Taking D_1, D_2 sufficiently large in terms of

q, K_0 , and then Γ sufficiently small compared with h , makes the right side at most CR^{-c} for some $c > 0$, uniformly in the component.

For later reference, put $G_{R,y} = G_{R,i,Q_0(y)}$ on the component X_i , and zero on the discarded pin remainder. Multiplying by the component weights and summing proves

$$\int \|(d_y)_*(P_R\mu - G_{R,y})\|_1 d\nu(y) \leq CR^{-c'} \quad (c' > 0). \quad (33)$$

On the discarded remainder use $\|P_R\mu\|_1 \leq C_T$ and its mass at most $R^{-\epsilon}$. This uses the first norm of the whole annular multiplier, rather than a sum of absolute packet norms there.

Initial reconstruction and the exact inherited identity

Fix a circle radius $r \asymp_T R$, and let ω_r be its normalized arclength measure. Choose a nonnegative compactly supported smooth frequency function ψ whose inverse transform is bounded away from zero on the fixed pin region. For every terminal Fourier label ℓ , with multiplier $\psi_{R,\ell}$ specified by the tree above, set

$$F_\ell = ((\psi_{R,\ell}\widehat{\mu}d\omega_r) * \psi)^\vee.$$

The terminal label is a standard label if its width equals δ_* , and a pair (S, I) otherwise. The fixed frequency convolution makes these functions smooth, even when the fine angular partition uses indicators. Define coarser functions by summing terminal descendants. In particular, summing all terminal descendants of S gives the original standard-cap function exactly.

On Q_0 , the circular extension of the retained standard packets, multiplied by $\check{\psi}$, agrees up to a rapid error with the sum of the retained full standard-cap functions. To verify this statement, sum all packets of each standard label first. A packet with $2T \cap Q_0 = \emptyset$ has rapidly small circular extension on Q_0 , by the standard integration-by-parts packet estimate used in Liu's Lemma 4.2. Therefore retaining those remote packets from a bad standard label changes the extension there by only a rapid error.

The fixed source cutoff does not change this conclusion. The kernel of a smooth standard annular-cap multiplier is concentrated at transverse scale $R^{-1/2}$ and longitudinal scale R^{-1} . Outside a fixed neighborhood of $\text{supp } \mu$, that multiplier applied to μ , together with any prescribed finite number of derivatives, is rapidly small in integrable norms. Consequently replacing $\chi(\psi_{R,S}\widehat{\mu})^\vee$ by $(\psi_{R,S}\widehat{\mu})^\vee$ changes its circular extension by a rapid error, uniformly for the radii in question. All source functions used for pinned densities still include χ , preserving their fixed compact support and source-pin separation.

All terminal descendants of a standard label have the same survival decision at Q_0 . Thus, only after this whole-standard-cap reconstruction, partition each retained standard spectral measure into its terminal Fourier descendants. It follows that

$$\check{\psi}(y)(G_{R,i,Q_0} * \widehat{\omega}_r)(y) = F_{Q_0,\text{root}}(y) + \text{RapDec}(R), \quad y \in Q_0, \quad (34)$$

where the function on the right is the sum of surviving terminal labels. No fine angular cutoff has been applied to an individual source packet. In particular, no localization assertion about the spatial tails of such an individual fine piece is needed.

More generally, for an active level- j cube and a parent angular label α , define $F_{Q_j,\alpha}$ to be the sum of terminal descendants of α with no inherited bad mark at levels $j, j+1, \dots, K-1$. Level zero has a single parent label, the root. At level K , no marks remain, so $F_{Q_K,\ell} = F_\ell$. The level- j mark is constant on each level- j Fourier label; future marks are inherited from the parent cube. Hence

$$F_{Q_j,\alpha} = \sum_{\substack{\theta \prec \alpha \text{ at level } j \\ \theta \text{ good for } Q_j}} F_{Q_{j+1},\theta}. \quad (35)$$

This is an exact partition of terminal labels. The function on the right for a fixed parent and label is independent of the child cube. After the standard angular scale, every future mark depends only on the standard parent S ; there a fixed fine function is simply retained or omitted as a whole. This is compatible with the exact identity.

One threshold factor per inflation step

Let m_Q be normalized Lebesgue measure on Q , and define

$$\mathcal{E}_j = \sum_{Q_j \text{ active}} \sigma_i(Q_j^+) \sum_{\alpha \text{ at level } j-1} \int |F_{Q_j, \alpha}|^2 dm_{Q_j^+}.$$

The level-zero angular sum consists only of the root. Initial localization and (34) give

$$\int |G_{R, i, Q_0(y)} * \hat{w}_r(y)|^2 d\sigma_i(y) \leq C_T R^{C\zeta + C_{K_0}h} \mathcal{E}_0 + \text{RapDec}(R).$$

Indeed each initial Fourier function has frequencies of size $O_T(R)$. Its pointwise square is controlled by a normalized average on an R^{-1} -ball with rapidly decreasing tails. Replacing that ball by an enlarged initial cube costs at most its area ratio $R^{2\zeta + O(T/N) + O_{K_0}(h)}$. Multiplication by its pin cube mass and summation give the displayed bound without any regularity assumption on the conditional pin measure.

We prove

$$\mathcal{E}_j \leq C_T R^{C_{K_0}h} H_{i,j} \mathcal{E}_{j+1} + \text{RapDec}(R). \quad (36)$$

Apply Proposition D.1 to each spatial parent and parent angular label, with $L = R^h$, $a = a_j$, $b = a_{j+1}$, $\sigma = \sigma_i$, and $H = H_{i,j}$. Its hypotheses are verified in detail at the end of Appendix D: admissibility controls the curvature of the frequency arc, the coarsened marking controls its directional mismatch, and the same enlarged parent normalizes both deletion and embedding. Summing the proposition over parents and labels gives (36). Its error is rapidly decreasing because all relevant counts and global norms are polynomial in R , while the decay order in the proposition is arbitrary. This step introduces one factor $H_{i,j}$, independently of the number of finer angular labels.

The terminal estimate and the chain cost

Iterating (36), using (30) and (29), gives the multiplicative loss

$$R^{C + C\zeta + C_{q, K_0}h + CK\Gamma}.$$

At level K , the spatial scale is one and the functions are the terminal Fourier functions F_ℓ with no remaining marks. Their global Lebesgue norms bound the remaining unit-scale averages, up to the allowed enlargement factors. By Plancherel and Cauchy–Schwarz,

$$\sum_\ell \|F_\ell\|_2^2 \leq C_T R^{-1} \int_{S^1} |\hat{\mu}(r\omega)|^2 d\omega. \quad (37)$$

For clarity, the normalized mass of the circle in a fixed-radius frequency ball is $O_T(R^{-1})$. For each terminal label, apply Cauchy–Schwarz to the convolution defining F_ℓ , using this mass bound. Integrate in frequency and sum the squared multipliers; their bounded overlap proves (37). This reasoning is valid for all terminal widths $\delta_{K-1} \leq R^{-1/2}$, including strictly finer widths. It does not introduce their number as a factor.

We conclude

$$\int |G_{R,i,Q_0(y)} * \widehat{\omega}_r(y)|^2 d\sigma_i(y) \leq C_T R^{-1+\mathcal{C}+C\zeta+C_{q,K_0}h+CK\Gamma} \int_{S^1} |\widehat{\mu}(r\omega)|^2 d\omega + \text{RapDec}(R). \quad (38)$$

Together with (33), this supplies the two shell estimates needed in the following summation argument. Its parameter choices may be made in the same order: first ζ , fixing K_0 ; then the threshold constants in terms of q, K_0 ; then h ; and finally T, ϵ , making Γ sufficiently small. Fixed constants depending on the resulting annular ratio are absorbed for sufficiently large R .

In particular the proved transfer uses the cost of the unforced chain (30). The source packet decomposition remains at the standard scale throughout; only the spectral functions used for quadratic inflation are refined past that scale.

B.4 Summation and identification of the distance law

Fix $\eta > 0$ with $\mathcal{C} + \eta < s - 1$. Choose ζ so that $C\zeta < \eta/4$, fixing a bound for K . Next choose the constants D_1, D_2 in (29). Choose h small enough that all $C_{q,K}h$ losses are below $\eta/4$, and finally choose T large and ϵ small so that $\Gamma \ll h$ and $CK\Gamma < \eta/4$. These choices are fixed independently of R . Logarithms and fixed T, K constants are absorbed for sufficiently large R .

For $y \in X_i$ set $G_{R,y} = G_{R,i,Q_0(y)}$, and on the discarded pin remainder set $G_{R,y} = 0$. Equation (32) and the parameter choices give

$$\int \|(d_y)_*(P_R\mu - G_{R,y})\|_1 d\nu(y) \leq CR^{-c}, \quad c > 0. \quad (39)$$

On the remainder, use the uniformly bounded first norm of the *whole annular multiplier*; push-forward decreases total variation, so its contribution is at most $C_T 2^{-\epsilon N}$. One must not estimate it by the sum of absolute values of all packets.

Liu's pinned quadratic identity [4] bounds the radial density's squared norm by a constant times

$$\int_0^\infty |G_{R,y} * \widehat{\omega}_r(y)|^2 r dr.$$

The source-pin distances lie in a fixed interval bounded away from zero, so its radial weights are equivalent to the unweighted first-dimensional density norm. The identity is applied separately at each pin and therefore permits the dependence of $G_{R,y}$ on y .

Here is an explicit uniform bound for the omitted frequencies. A standard source packet has the form $\chi\eta_T(\psi_{R,S}\widehat{\mu})^\vee$, with $\|D^L(\chi\eta_T)\|_1 \leq C_L R^{L/2}$. Outside a fixed enlarged annulus, the distance to every $\text{supp } \psi_{R,S}$ is comparable to $R + |\xi|$. Integrating the convolution in frequency, using $|\widehat{\mu}| \leq 1$ and the polynomial number of packets, gives

$$|\widehat{G}_{R,y}(\xi)| \leq C_L R^{C+L/2} (R + |\xi|)^{-L}.$$

The constant C is independent of L , of the pin, and of all packet selections. Since ω_r is normalized, the corresponding omitted circular energy is at most

$$C_L R^{2C+L} \int_0^\infty (R+r)^{-2L} r dr \leq C_L R^{2C+2-L}.$$

Taking L arbitrarily large justifies the rapidly decreasing error uniformly over all components and pins. Integrating (38) over the remaining annulus and using

$$\int_{|\xi| \asymp_T R} |\widehat{\mu}(\xi)|^2 d\xi \leq C_T R^{2-s} I_s(\mu)$$

now gives

$$\int \|(d_y)_* G_{R,y}\|_2^2 d\nu(y) \leq C R^{1-s+C+\eta} I_s(\mu). \quad (40)$$

To combine components, multiply their bounds by w_i and sum. The sets X_i are disjoint, and $\sum_i w_i \leq 1$; no factor counting the components appears.

The exponents in (39) and (40) imply convergence of the bad shell densities in $L^1(d\nu dt)$, and of the good shell densities in $L^2(d\nu dt)$. All source functions have the same fixed compact support, so the radial variable ranges over a bounded interval. Include the smooth low-frequency density. Their sum is an integrable joint density.

To identify it, finite shell sums reconstruct μ as a distribution. If $\Phi(y, t)$ is a smooth test function, then

$$x \longmapsto \chi(x) \int \Phi(y, |x - y|) d\nu(y)$$

is a smooth compactly supported test function: the cutoff χ keeps source and pins separated. Distributional reconstruction, followed by the stated norm convergence, identifies the joint density with the measure induced by $(y, x) \mapsto (y, |x - y|)$ on $\nu \times \mu$. Uniqueness of disintegration proves (18). In particular these pinned probabilities are nonzero and supported on the corresponding compact distance sets.

C Positive approximation below the bound $2d - 1$

This section proves the first branch directly. The approximating measures are positive probabilities throughout, and converge in first norm to the actual pinned distance law. The only specialized external input is the following published radial-projection theorem.

Theorem C.1 (Orponen's averaged radial-projection theorem). *Let $\alpha > 1$. For two compactly supported planar probabilities μ, ν with disjoint supports and bounds*

$$\mu(B(z, r)) + \nu(B(z, r)) \leq Cr^\alpha, \quad 0 < r \leq 1,$$

there is a number $q = q(\alpha) > 1$ such that, for μ -almost every x ,

$$(\Theta_x)_* \nu = \rho_x \sigma, \quad \Theta_x(y) = \frac{x - y}{|x - y|}, \quad B := \int \|\rho_x\|_{L^q(\sigma)}^q d\mu(x) < \infty. \quad (41)$$

Here σ is normalized arclength on the circle.

This is the form of Orponen's result [1] stated in [3, Theorem 3.7]. A common translation and dilation puts the supports in the unit disk required there. Reversing the radial direction does not change an angular norm. In particular, different Frostman exponents $s, t > 1$ are allowed: choose $1 < \alpha < \min\{s, t\}$.

C.1 A projection estimate that permits correlated angles

We give the analytic estimate in detail, since the two directions used later are functions of the same pin. The Fourier convention is $\widehat{\lambda}(\xi) = \int e^{-2\pi i x \cdot \xi} d\lambda(x)$.

Lemma C.2. *Fix $1 < s < 2$ and geometric constants. Let $0 < r \leq 1$, and let λ be a probability supported in $B(0, C_0 r)$, with $\lambda(B(z, t)) \leq Kt^s$ for all $z, t > 0$. Let $F(\theta)$ be the density of the orthogonal projection $x \mapsto x \cdot \theta$, defined for almost every θ . Suppose a finite positive measure κ carries measurable directions θ, ϕ and a scalar shift v , with both angular marginals bounded by $A\sigma$, and*

$$d(\theta, \phi) \leq C_1 r, \quad |v| \leq C_1 r^2.$$

Then

$$\int \|F(\theta)(\cdot - v) - F(\phi)\|_2^2 d\kappa \leq CAKr^{2s-2}. \quad (42)$$

No independence of directions or shifts is assumed.

Proof. For a frequency radius $R > 0$, the Gaussian Fourier identity gives

$$\begin{aligned} E_R &:= \int e^{-|\xi|^2/R^2} |\widehat{\lambda}(\xi)|^2 d\xi \\ &= cR^2 \iint e^{-c'R^2|x-x'|^2} d\lambda(x) d\lambda(x') \leq C_s K R^{2-s}. \end{aligned} \quad (43)$$

The last bound follows by the Frostman estimate on the ball of radius R^{-1} and on the successive doubled annuli. Positivity of the spatial Gaussian also proves, for the signed coordinate measures,

$$\int e^{-|\xi|^2/R^2} |\widehat{x_k \lambda}(\xi)|^2 d\xi \leq Cr^2 E_R, \quad k = 1, 2: \quad (44)$$

in the spatial double integral, the additional factor has absolute value $|x_k x'_k| \leq C_0^2 r^2$.

Choose smooth scalar-frequency cutoffs ψ_R , $R \in 2^{\mathbb{Z}}$, supported where $R/2 \leq |\tau| \leq 2R$, so that $\sum_R \psi_R(\tau)^2 = 1$ for $\tau \neq 0$. Put

$$U_R(\theta) = \mathcal{F}_\tau^{-1}[\psi_R(\tau)\widehat{\lambda}(\tau\theta)].$$

Polar coordinates and (43) give

$$\int \|U_R(\theta)\|_2^2 d\sigma(\theta) \leq CKR^{1-s}, \quad \int \|\partial_\theta U_R(\theta)\|_2^2 d\sigma(\theta) \leq CKr^2 R^{3-s}. \quad (45)$$

For the derivative bound use (44): differentiating $\widehat{\lambda}(\tau\theta)$ inserts a bounded linear combination of $\widehat{\tau x_k \lambda}(\tau\theta)$. The projected densities exist and are jointly square integrable in angle and scalar variable. Indeed, summing the first estimate over $R \geq 1$ converges because $s > 1$; for $R < 1$ use $|\widehat{\lambda}| \leq 1$, giving instead the summable bound CR . Plancherel then identifies the projected measures with these densities outside one angular null set.

The fundamental theorem of calculus for the smooth Hilbert-valued function U_R , along the shorter angular arc, gives

$$\|U_R(\theta) - U_R(\phi)\|_2^2 \leq Cr \int_{d(z, \theta) \leq C_1 r} \|\partial_z U_R(z)\|_2^2 dz.$$

Integration with the first marginal and Fubini contribute another factor Ar . The trivial squared-difference bound uses both marginals. Together with (45) these yield

$$\int \|U_R(\theta) - U_R(\phi)\|_2^2 d\kappa \leq CAK \min\{1, (r^2 R)^2\} R^{1-s}.$$

For translations, the pointwise scalar Fourier multiplier bound

$$|e^{-2\pi i \tau v} - 1|^2 \leq C \min\{1, (r^2 R)^2\}, \quad |\tau| \asymp R,$$

gives the same estimate. It is valid before integrating in the pin parameter, so the shift may be correlated with either direction.

For each parameter outside the angular marginal null sets, Plancherel and the squared partition of unity express the squared norm of the full difference as the sum of its band squared norms. The squared triangle inequality and Tonelli therefore reduce the assertion to

$$\sum_{R \in 2^{\mathbb{Z}}} \min\{1, (r^2 R)^2\} R^{1-s} \leq C_s r^{2s-2}.$$

Below $R = r^{-2}$ the summand is $r^4 R^{3-s}$; above it the summand is R^{1-s} . Both geometric sums have the stated bound. $\square$

C.2 Convergence of one fixed family of positive approximations

Proposition C.3. *Let μ, ν be compactly supported planar probabilities with positive separation. Suppose μ is s -Frostman, where $1 < s < 2$, and ν has a Frostman exponent greater than one. If the number N_n of occupied source dyadic squares of side 2^{-n} satisfies*

$$N_n \leq C2^{un}, \quad u < 2s - 1, \quad (46)$$

then $(d_y)_ \mu$ is absolutely continuous for ν -almost every y .*

Proof. Apply Theorem C.1. Write $M(x) = \|\rho_x\|_q^q$, so $\int M d\mu = B < \infty$. For each positive-mass half-open dyadic square Q , put $p_Q = \mu(Q)$ and $\mu_Q = p_Q^{-1}\mu|_Q$. Choose once and for all a point $x_Q \in Q \cap \text{supp } \mu_Q$ where the radial density exists and

$$M(x_Q) \leq \frac{2}{p_Q} \int_Q M d\mu. \quad (47)$$

Such points exist by averaging. There are only countably many nodes, and at every level

$$\sum_{Q \text{ at level } n} p_Q M(x_Q) \leq 2B. \quad (48)$$

For $x \in Q$, define

$$L_Q(x, y) = \frac{x_Q - y}{|x_Q - y|} \cdot (x - y).$$

Let f_n be the density of the pushforward of $\mu \times \nu$ by $(x, y) \mapsto (y, L_{Q_n(x)}(x, y))$, relative to $\nu(dy)dt$. These densities exist: each conditional source has finite first energy, its orthogonal projections have square-integrable densities for almost every direction, and the chosen center's radial pin law is absolutely continuous. The countably many conditional densities can be chosen jointly measurably. Thus

$$f_n \geq 0, \quad \int f_n d\nu dt = 1.$$

Positive separation and Taylor's theorem give, uniformly for source and pin points and sufficiently large n ,

$$|L_{Q_n(x)}(x, y) - |x - y|| \leq C2^{-2n}. \quad (49)$$

We compare consecutive levels. Put $r = 2^{-n}$, let Q be a child of P , and apply both affine maps to the same μ_Q . Recenter this source at $b = x_Q$. With $\theta = \Theta_{x_P}(y)$ and $\phi = \Theta_b(y)$,

$$L_P(x, y) = \theta \cdot (x - b) + \theta \cdot (b - y), \quad L_Q(x, y) = \phi \cdot (x - b) + |b - y|.$$

The directions differ by $O(r)$, while the constants differ by

$$|\theta \cdot (b - y) - |b - y|| = |b - y| |\theta \cdot \phi - 1| \leq Cr^2. \quad (50)$$

The recentering is what gives this quadratic error.

Retain only pins where both selected angular densities are at most $A \geq 1$. The two retained angular marginals are bounded by $A\sigma$, and the discarded pin mass is at most

$$A^{1-q}(M(x_P) + M(x_Q)).$$

The translated conditional source is supported in a ball of radius Cr and has Frostman constant at most C_μ/p_Q . Lemma C.2 gives retained integrated squared second-norm error at most

$$\frac{CA}{p_Q} r^{2s-2}.$$

For each pin, the two conditional densities occupy the union of two intervals of total length Cr . Cauchy–Schwarz, first on that union and then over retained pins, bounds their integrated first-norm difference by $CA^{1/2}p_Q^{-1/2}r^{s-1/2}$. On discarded pins use the bound two for the difference of probability densities.

Multiply by p_Q and sum. The retained terms satisfy $\sum_Q \sqrt{p_Q} \leq \sqrt{N_{n+1}}$; the discarded parent terms sum with the parent masses, because the child masses sum to their parent's mass. Equation (48) therefore gives

$$\|f_{n+1} - f_n\|_{L^1(\nu \times dt)} \leq CA^{1/2}r^{s-1/2}\sqrt{N_{n+1}} + CBA^{1-q} \leq C\sqrt{A}r^{(2s-1-u)/2} + CBA^{1-q}. \quad (51)$$

No new centers are chosen for this comparison.

Write $\kappa = 2s - 1 - u > 0$ and choose $A = r^{-\kappa/2}$. The last bound is

$$Cr^{\kappa/4} + CB r^{\kappa(q-1)/2},$$

which is summable over dyadic scales. Hence the same positive probabilities f_n converge in first norm to a nonnegative density of mass one. Equation (49) identifies their weak limit as the pushforward of $\mu \times \nu$ by $(x, y) \mapsto (y, |x - y|)$. The first-norm limit is this same measure. Uniqueness of disintegration gives its density on almost every pin fiber, proving the proposition. $\square$

This proves Proposition 4.2: a bounded t -Frostman probability is also s -Frostman when $s < t$, and an $O(r^{-u})$ ball-covering bound gives $O(2^{un})$ occupied dyadic squares by bounded overlap at comparable scales. The compact reduction needed for the Borel theorem has already been proved in Lemma 4.1.

D A weighted embedding for one Fourier step

This appendix proves the one-step estimate (36) directly from its geometric and Fourier hypotheses. In particular, the selected caps may depend arbitrarily on the child cube. The proof does not restore the omitted caps before applying the good-tube condition. The pin measure need not satisfy a Frostman bound in this step.

For a bounded rectangle A of positive area, write $m_A = |A|^{-1} \mathcal{L}^2|_A$. Dilations of cubes and rectangles below are concentric. We use the Fourier convention $\widehat{f}(\xi) = \int e^{-2\pi i x \cdot \xi} f(x) dx$.

Proposition D.1 (Selected-cap weighted embedding). *Fix an integer $K \geq 1$, an index $0 \leq j < K$, and constants $C_0, B_0 \geq 1$. Let $0 < a \leq b$, and let P be an axis-parallel square of side b . Let $\mathcal{Q}$ be a finite subfamily of squares of an axis-parallel a -grid, each contained in P . For $L \geq L_0(C_0, K)$, set*

$$Q^+ = L^{2j+2}Q, \quad \widetilde{Q} = LQ^+, \quad P^+ = L^{2j+4}P, \quad D = L^{4K+20}.$$

Let σ be a finite positive Borel measure, and put

$$w_Q = \sigma(Q^+), \quad M = \sigma(P^+).$$

Let Θ be a finite label set. For every $\theta \in \Theta$ choose a function $f_\theta \in L^2(\mathbb{R}^2)$, a unit vector u_θ , a frequency center ξ_θ , and an effective unit direction v_θ , subject to the following conditions:

1. The Fourier support of f_θ is contained in

$$\left\{ \xi : |(\xi - \xi_\theta) \cdot u_\theta^\perp| \leq C_0/a, \quad |(\xi - \xi_\theta) \cdot u_\theta| \leq C_0/b \right\}.$$

These supports are also contained in balls of radius C_0/a whose pointwise overlap is at most B_0 .

2. The angular mismatch obeys

$$|u_\theta - v_\theta| \leq C_0 a/b.$$

Directions may first be given consistent signs, since the rectangles and the tests depend only on the unoriented lines.

3. For each child there are numbers $\varepsilon_{Q,\theta} \in \{0,1\}$. Let $T_{Q,\theta}$ be the rectangle centered at the center of Q , of width Da perpendicular to v_θ and length Db parallel to v_θ . For a number $H \geq 1$, assume only that

$$\varepsilon_{Q,\theta} = 1 \implies \sigma(T_{Q,\theta} \cap P^+) \leq H \frac{a}{b} M. \quad (\text{G})$$

The function f_θ is the same for all children; only the selection numbers depend on Q . Then, for every $m > 0$,

$$\begin{aligned} & \sum_{Q \in \mathcal{Q}} w_Q \int_{Q^+} \left| \sum_{\theta \in \Theta} \varepsilon_{Q,\theta} f_\theta \right|^2 dm_{Q^+} \\ & \leq CL^{8j+14} HM \sum_{\theta \in \Theta} \int_{P^+} |f_\theta|^2 dm_{P^+} + C_m M a^{-2} L^{-m} \sum_{\theta \in \Theta} \|f_\theta\|_2^2. \end{aligned} \quad (52)$$

Here C and C_m may depend on the fixed constants C_0, B_0, K , and C_m also on m . They do not depend on a, b, L, H , the measure, or the numbers of children and labels.

Proof. We separate the geometric consequence of the selected tests, local orthogonality, and the weighted reproducing-kernel estimate.

Geometry and the selected tube weights. The enlarged child cubes have overlap at most

$$\Lambda \leq CL^{4j+4}. \quad (53)$$

Indeed, in each coordinate at most $L^{2j+2} + 2$ grid centers can place a point in the corresponding enlarged interval. Also, for a point in $\tilde{Q}$, its distance from the center of P in the maximum norm is at most $(1 + L^{2j+3})b/2$. It follows that

$$\text{dist} \left(\bigcup_{Q \in \mathcal{Q}} \tilde{Q}, \mathbb{R}^2 \setminus P^+ \right) \geq Lb \quad (54)$$

for $L \geq 4$. In particular all Q^+ lie in P^+ , and

$$\sum_Q w_Q \leq \Lambda M. \quad (55)$$

Fix θ , and let U be any translate of a rectangle of width a and length b , oriented along u_θ . We claim

$$\sum_{\substack{Q: \varepsilon_Q, \theta=1 \\ \tilde{Q} \cap U \neq \emptyset}} w_Q \leq \Lambda H \frac{a}{b} M. \quad (56)$$

If the index set is empty this is immediate. Otherwise choose one of its cubes, Q_0 . Every Q^+ in that sum lies inside $T_{Q_0, \theta}$, as we now check.

Let q, q_0 be the centers of Q, Q_0 , and choose $z \in \tilde{Q} \cap U$, $z_0 \in \tilde{Q}_0 \cap U$. For $x \in Q^+$, write

$$x - q_0 = (x - q) + (q - z) + (z - z_0) + (z_0 - q_0).$$

The first, second, and fourth terms have length at most $CL^{2j+3}a$. The third has components at most a and b perpendicular and parallel to u_θ . Therefore

$$|(x - q_0) \cdot u_\theta^\perp| \leq CL^{2j+3}a, \quad |(x - q_0) \cdot u_\theta| \leq CL^{2j+3}b.$$

Changing to v_θ adds at most $CC_0L^{2j+3}a$ to the transverse component. The longitudinal component stays bounded by $C(1 + C_0)L^{2j+3}b$. Since $2j + 3 \leq 2K + 1$, the prescribed value of D exceeds twice these factors once $L \geq L_0(C_0, K)$. This proves the containment. All the cubes in question also lie in P^+ . Their overlap bound and the selected test at Q_0 now give

$$\sum_{\substack{Q: \varepsilon_Q, \theta=1 \\ \tilde{Q} \cap U \neq \emptyset}} \sigma(Q^+) \leq \Lambda \sigma(T_{Q_0, \theta} \cap P^+) \leq \Lambda H(a/b)M,$$

as claimed. Thus the mass bound for *every* dual rectangle in (56) is a consequence of the given tests at selected children, not an additional assumption.

Local orthogonality for a selected subset. Choose a Schwartz function v whose Fourier transform is supported in the unit ball and whose modulus is at least one on the unit square centered at zero. To obtain one, start with the inverse transform of a smooth bump supported in a sufficiently small ball; its value can be made nearly constant on that square and then multiplied by a fixed constant. For a child centered at q , put

$$v_Q(x) = v \left(\frac{x - q}{L^{2j+2}a} \right).$$

The product $v_Q f_\theta$ has Fourier support in the corresponding frequency ball enlarged by at most a^{-1} . These enlarged balls have overlap at most a fixed multiple of B_0 . In fact, dilating equal-radius balls by two increases their overlap by at most a factor nine in the plane: balls contributing at a point have their original balls inside the concentric ball of three times the radius, and an integration of the original overlap bound gives this count.

For every subset $A \subseteq \Theta$, Plancherel and pointwise Cauchy–Schwarz in frequency therefore give

$$\left\| v_Q \sum_{\theta \in A} f_\theta \right\|_2^2 \leq C \sum_{\theta \in A} \|v_Q f_\theta\|_2^2. \quad (57)$$

The constant is independent of the subset. Apply this to $A = \{\theta : \varepsilon_{Q,\theta} = 1\}$. Restrict the left side to Q^+ , and truncate the right side at $\tilde{Q} = LQ^+$. Schwartz decay and normalization of the cube averages give, for every $p > 0$,

$$\begin{aligned} \int_{Q^+} \left| \sum_{\theta} \varepsilon_{Q,\theta} f_\theta \right|^2 dm_{Q^+} \\ \leq CL^2 \sum_{\theta: \varepsilon_{Q,\theta}=1} \int_{\tilde{Q}} |f_\theta|^2 dm_{\tilde{Q}} + C_p a^{-2} L^{-p} \sum_{\theta} \|f_\theta\|_2^2. \end{aligned} \quad (58)$$

For the first term the area ratio is exactly L^2 . For the second, $|Q^+|^{-1} \leq a^{-2}$, and the square of the cutoff is bounded by $C_p L^{-p}$ outside $\tilde{Q}$. No unselected label has been introduced into the main term.

The anisotropic weighted embedding. For a fixed label define the finite positive averaging measure

$$\tau_\theta = \sum_{Q: \varepsilon_{Q,\theta}=1} w_Q m_{\tilde{Q}}.$$

Equations (56) and (55) imply

$$\tau_\theta(U) \leq \Lambda H(a/b)M \quad \text{for every dual rectangle } U, \quad \tau_\theta(\mathbb{R}^2) \leq \Lambda M. \quad (59)$$

Take a smooth compact frequency bump equal to one on $[-C_0, C_0]^2$. Translating, rotating, and rescaling it produces a reproducing kernel k_θ for the Fourier rectangle in the proposition. Thus $f_\theta = f_\theta * k_\theta$, and

$$\begin{aligned} \|k_\theta\|_1 &\leq C, \\ |k_\theta(z)| &\leq \frac{C_p}{ab} \left(1 + \frac{|z \cdot u_\theta^\perp|}{a} + \frac{|z \cdot u_\theta|}{b} \right)^{-p} \quad (p > 0). \end{aligned} \quad (60)$$

The translation of the frequency rectangle merely modulates the kernel and does not change these absolute-value estimates. Each L^2 function with the specified bounded Fourier support has a continuous representative, since its Fourier transform is also integrable. The reproducing identity and the following pointwise estimate are understood for that representative. Cauchy–Schwarz yields

$$|f_\theta(x)|^2 \leq C \int |f_\theta(y)|^2 |k_\theta(x-y)| dy.$$

For each fixed y , partition the plane into translates of dual rectangles a -by- b centered on the corresponding rectangular lattice based at y . The mass estimate in (59), the kernel bound with any $p > 2$, and summation of the lattice decay give

$$\int |k_\theta(x-y)| d\tau_\theta(x) \leq \frac{C}{ab} \Lambda H \frac{a}{b} M = C \frac{\Lambda H M}{b^2}. \quad (61)$$

This computation displays the single occurrence of the tube threshold.

If $y \notin P^+$, the support margin (54) gives $|x - y| \geq Lb$ on the support of τ_θ . Since $a \leq b$, the anisotropic distance in (60) is then at least a fixed multiple of L . Its arbitrary-order decay and the total mass in (59) give the additional bound

$$\int |k_\theta(x - y)| d\tau_\theta(x) \leq C_p \frac{\Lambda M}{ab} L^{-p} \quad (y \notin P^+).$$

Integrating the pointwise estimate for f_θ , and using Tonelli's theorem, now proves

$$\int |f_\theta|^2 d\tau_\theta \leq C \frac{\Lambda HM}{b^2} \int_{P^+} |f_\theta|^2 dy + C_p \frac{\Lambda M}{ab} L^{-p} \|f_\theta\|_2^2. \quad (62)$$

Finally sum (58) with weights w_Q , and apply (62) to each label. The main coefficient, after converting to the normalized parent average, is

$$CL^2 \Lambda HM \frac{|P^+|}{b^2} \leq CL^{2+4j+4+4j+8} HM = CL^{8j+14} HM.$$

The local-orthogonality remainder is bounded by $C_p \Lambda M a^{-2} L^{-p} \sum_\theta \|f_\theta\|_2^2$. The weighted-kernel remainder has the same bound with an additional L^2 , since $(ab)^{-1} \leq a^{-2}$. Choosing the arbitrary decay order p larger than $m + 4j + 6$, and absorbing (53), gives the stated remainder. This proves the proposition. $\square$

Verification of the hypotheses in the Fourier argument

Fix a spatial parent $P = Q_{j+1}$ and one parent angular label in Section B.3. Apply Proposition D.1 with

$$a = a_j = r_j/R, \quad b = a_{j+1} = r_{j+1}/R, \quad L = R^h, \quad \sigma = \sigma_i, \quad H = H_{i,j}.$$

The spatial dilations and the test dilation then coincide exactly with those specified there, with $K = K_0$. In particular the proposition does not require replacing the current tests by tests on additional parents or by a new measure.

Take

$$f_\theta = F_{P,\theta}, \quad \varepsilon_{Q,\theta} = \mathbf{1}_{\{\theta \text{ is good for } Q\}}.$$

The inherited identity (35) says that each child's function is exactly the selected sum in the proposition. For this fixed parent the functions $F_{P,\theta}$ do not depend on the child. The selections are constants on the averaging cubes, even though different cubes may choose completely different cap subsets.

The actual Fourier direction u_θ is the direction of the auxiliary angular interval, and v_θ is its effective marking direction. Their difference is $O(\bar{\delta}_j)$, with $\bar{\delta}_j \leq a_j/a_{j+1}$ by (31). This supplies the angular mismatch hypothesis. The marking rule supplies (G): multiplying its conditional probability bound by $\sigma_i(P^+)$ gives precisely the unnormalized inequality there. Thus the stronger-looking bound (56) for arbitrary dual rectangles has been derived from the existing marks, rather than silently assumed.

The circular support of each auxiliary cap has tangential frequency width $O_T(R/r_j) = O_T(a^{-1})$ and normal width $O_T(R/r_j^2) \leq O_T(R/r_{j+1}) = O_T(b^{-1})$. The fixed frequency convolution adds bounded width, absorbed because $b \leq 1$. The supports lie in bounded-overlap balls of radius $O_T(a^{-1})$. When labels are pairs (S, I) , only boundedly many standard labels S are nonzero at a direction, and the dyadic intervals have bounded overlap. Their enclosing balls have uniformly bounded overlap as well: consecutive angular centers at a given level are separated by a fixed

multiple of δ_j , up to this bounded multiplicity. The constants C_0, B_0 may depend on the fixed annular ratio, hence on T , but not on R . The terminal widths are at least a constant times R^{-1} , so the fixed frequency smoothing causes no growing ball multiplicity.

All functions in this application belong to L^2 . For example, a bounded-density finite circular measure convolved with the fixed compact smooth frequency function has a smooth compactly supported Fourier representative. This observation applies also to the hard angular indicator pieces; no differentiation of those indicators is needed anywhere in Proposition D.1.

Consequently (52), followed by summation over parents and parent angular labels, proves (36), with a loss $R^{C_{\kappa_0}h}$ and the single factor $H_{i,j}$. The remainder is rapidly decreasing in the required sense. Indeed $a_j \geq R^{-1}$, and for any target decay power A , the remainder in the proposition contains

$$a_j^{-2} L^{-m} \leq R^{2-hm}.$$

For fixed $h > 0$, choose m sufficiently large to absorb A and the polynomial bounds on the remaining sums and global norms. Constants depending on this choice are independent of the frequency.

References

- [1] T. Orponen. On the dimension and smoothness of radial projections. *Analysis & PDE* 12 (2019), 1273–1294. [arXiv:1710.11053](#).
- [2] T. Keleti and P. Shmerkin. New bounds on the dimensions of planar distance sets. *Geometric and Functional Analysis* 29 (2019), 1886–1948. [arXiv:1801.08745v3](#). Lemmas 3.4 and 5.4, Corollary 3.5, Proposition 3.11.
- [3] L. Guth, A. Iosevich, Y. Ou and H. Wang. On Falconer’s distance set problem in the plane. *Inventiones Mathematicae* 219 (2020), 779–830. [arXiv:1808.09346v1](#).
- [4] B. Liu. An L^2 -identity and pinned distance problem. *Geometric and Functional Analysis* 29 (2019), 283–294. [arXiv:1802.00350v3](#), Theorem 1.9.
- [5] B. Liu. Lebesgue measure of distance sets with regular pins and multi-scale Mizohata–Takeuchi-type estimates. Version 3, 16 July 2026. [arXiv:2603.15328v3](#), Lemma 4.2.